

# Anahita Sharma

+91 7006072448 — sharma.anahita.as@gmail.com — github.com/sharma-anahita — LinkedIn — Portfolio

## Education

**National Institute of Technology, Srinagar**

B.Tech in Computer Science and Engineering

CGPA: **9.16 / 10**

Srinagar, J&K

**2023 – 2027**

## Experience

**Defence Research and Development Organisation (DRDO)**

Delhi, India

*Research Intern — Office of Cyber*

**Dec 2025 – Feb 2026**

- Analyzed low-level system behavior including process execution, memory management, and runtime failures to improve system reliability.
- Debugged multi-component system workflows by identifying failure patterns and root causes, improving issue resolution efficiency.
- Applied operating system concepts (process lifecycle, system calls, memory handling) to optimize execution flows and performance.

## Projects

**MindTrack — Scalable Mental Health Platform** — Live — GitHub    Node.js, Express.js, MongoDB, React (TypeScript)

- Architected a modular full-stack platform supporting user data, analytics, and engagement tracking with scalable backend services.
- Designed RESTful APIs for mood tracking and insights, improving request handling efficiency and response performance.
- Integrated a Python-based ML service for sentiment analysis to generate personalized insights.
- Developed a context-aware LLM-based conversational module to enhance user interaction and engagement.
- Optimized database performance using indexing and pagination to improve query efficiency.

**Cross-Platform Custom Shell** — GitHub

C, POSIX, Win32 API

- Developed a Unix-like shell supporting command execution, pipelines, and I/O redirection.
- Implemented process management and inter-process communication using POSIX and Win32 APIs.
- Designed command parsing and execution flow for efficient system-level operations.

**Backend Platform for Scalable Content Services** — GitHub

Node.js, Express.js, MongoDB

- Built scalable REST APIs for authentication, content delivery, and user management.
- Implemented secure authentication using JWT, refresh token rotation, and role-based access control.
- Optimized database queries using indexing and aggregation pipelines for efficient data retrieval.
- Ensured system reliability through structured error handling and asynchronous request processing.

**Gmail to Google Sheets Automation** — GitHub

Python, OAuth 2.0, Gmail API, Google Sheets API

- Built an automated pipeline to extract unread emails from Gmail and log structured data into Google Sheets.
- Implemented secure OAuth 2.0 authentication for Gmail and Sheets API access without storing credentials.
- Developed retry logic with exponential backoff and structured logging for reliable API interactions.
- Implemented email parsing with HTML-to-text conversion for clean and readable data storage.

## Skills

**Languages:** C++, Java, Python, JavaScript

**Core CS:** Data Structures & Algorithms, Operating Systems, Computer Networks

**Backend & Systems:** Node.js, REST APIs, Distributed Systems, Microservices, System Design

**Databases:** MongoDB, MySQL

**Tools:** Git, Linux, Postman

## Achievements

- Solved 1000+ Data Structures and Algorithms problems; LeetCode Rating **1836** (Top 6%).
- Flipkart GRiD 7.0 — **National Semi-Finalist** (2025).
- Postman API Fundamentals Student Expert.